

VERIFIED EXECUTABLE LINEAR ALGEBRA

# DenseMatrix makes Lean matrix algorithms run like data.

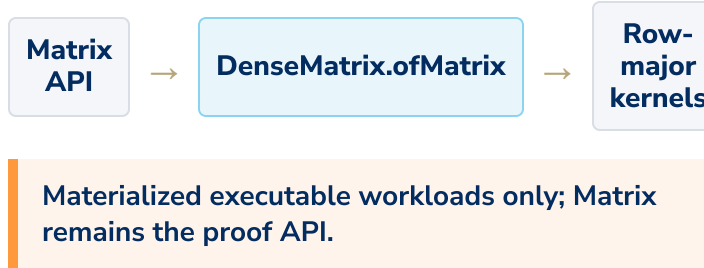
Row-major data for RREF, LU, and determinant kernels, bridged back to proof-friendly `Matrix` theorems.

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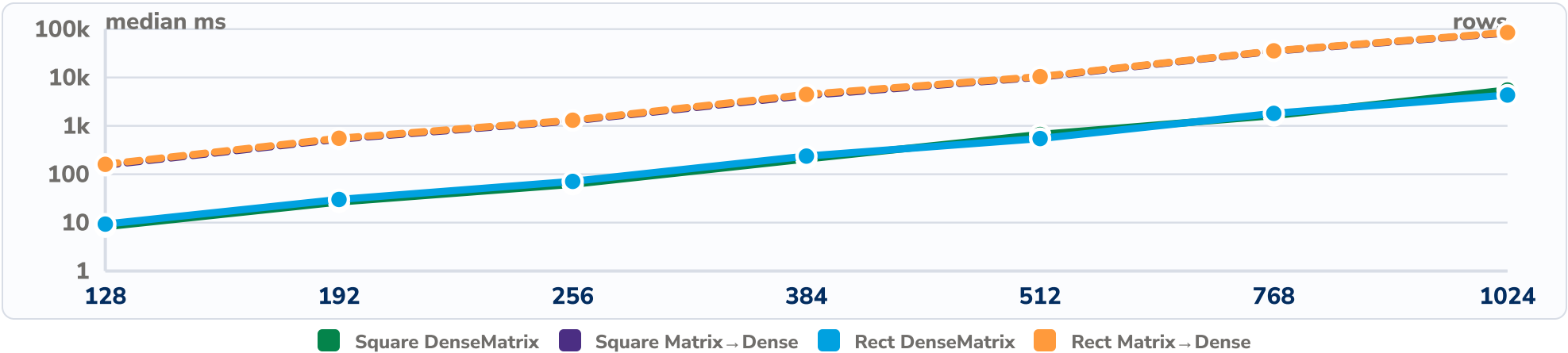
## Why DenseMatrix

Row-major arrays let kernels work over contiguous data instead of re-evaluating pointwise Matrix functions.



## Multiplication Scaling

DenseMatrix vs Matrix → Dense, median ms on a log scale.



## Proof Connection

- DenseMatrix.toMatrix\_add
- DenseMatrix.toMatrix\_mul
- rowEchelonForm\_toMatrix
- luDet\_eq\_det
- gaussDet\_eq\_det

Theorems connect executable kernels back to the Matrix API.

## Representation Tradeoff

Pointwise primitives at the largest complete size.

| OPERATION | DENSE | MATRIX | MATRIX → DENSE | WINNER |
|-----------|-------|--------|----------------|--------|
| Add       | 54.0  | 45.1   | 47.5           | Matrix |
| Scalar    | 53.2  | 30.3   | 35.3           | Matrix |
| Transpose | 55.3  | 43.5   | 46.8           | Matrix |

## Benchmark Scope

|              |                                    |               |
|--------------|------------------------------------|---------------|
| 2247 RECORDS | 328 MATERIALIZED ROWS              | 2048 MAX ROWS |
| Protocol     | 30 repeats · 5 warmups             |               |
| Sizes        | 16, 32, 64, 96, 128, 192, 256      |               |
| Stress       | 384, 512, 768, 1024                |               |
| Machine      | pinned core · environment archived |               |

## Algorithm Kernels

Representative Rat kernels at n=128.

|           |       |          |
|-----------|-------|----------|
| REF       | n=128 | 16292 ms |
| RREF      | n=128 | 32394 ms |
| LU factor | n=128 | 17090 ms |
| Gauss det | n=128 | 16234 ms |
| LU det    | n=128 | 16933 ms |